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Foamed Co2 Fracturing: Enhancing Productivity in Carbonate Formations in High Pressure High Temperature Wells in Middle East

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Abstract

The primary purpose of pumping foamed CO2 fluids during fracturing operations is to aid in post-fracturing flowback especially in tight and(or) low pressure formations, where reservoir is not supporting sufficient energy to flowback the fracturing fluids.

This type of treatment has shown to be beneficial in deep and tight carbonate formations together with a tailored set of acid systems. The pumping schedule was typically designed with three primary treatment fluids to be pumped in a series of cycles. These fluids include a linear gel, 28% HCl, and a viscoelastic self-diverting acid based on 15%-20% HCl. The target range for bottom hole foam quality was 55-65%. Since CO2 is pumped as a liquid, standard hydraulic fracturing pumps could be utilized. Vertical single stage, vertical multi-stage and horizontal multistage completions were successfully utilized during the multi-year CO2 fracturing campaign across one of the World's largest conventional fields.

This paper will present an evolution of the hi-tier CO2 pumping operations in the extreme geological environment of tight HPHT carbonate formations. Evolution of the CO2-based foam fracturing technology is provided with support from engineering and implementation teams throughout the campaign since the start of the first CO2 job. Significant progress has been made in well completions to accommodate the CO2 technology treating low pressure wells. Such wells once treated with normal fluids, nitrogen lifting is used resulting in additional operational efforts. Technology is particularly helpful in reducing the water consumption and polymer residues by replacing fluids with CO2 up to 60% per stage. This allowed to leave a significantly cleaner reservoir with additional energy to start hydrocarbons production faster, resulting in significant increase in wells productivity when compared to conventional fracturing method. Quality of the CO2 foam fracturing operations was under control without any issues occurred during the campaign (outperforming average conventional operations efficiency).

Comprehensive study that includes logistics, mixing and execution side of these treatments as well as engineering and field development aspects will be of vital interest to stimulation experts in conventional gas resources and extend this expertise to unconventional resources.

Introduction

Foamed CO_2 treatments provide effective solutions for hydrocarbon extraction in reservoirs experiencing significant pressure depletion (Briscoe, H. R. 1979); (Ford, W. G. F. 1981) . In the Middle East, where subterranean carbonate formations are prolific, foamed CO_2 acidizing has been used to target deep, high-temperature, sub-hydrostatic conditions within these formations (Malik, A. R.,2014);(Sanchez, M 2015); (Rahim, Z. Al-Kanaan 2018);(Abel, J. T 2019);(Gromakovskii, D 2019);(Almutary, T 2021);(Liu, P 2022); (Al-Muhanna, D 2022); (Basset, M 2024);(Asel, S 2024). A key advantage of this method is its ability to eliminate the need for post-treatment nitrogen lifting via coiled tubing intervention to initiate flowback. In conventional acidizing, nitrogen lifting is often required to initiate well kickoff in depleted reservoirs. The energized nature of the foam, primarily due to the internal phase of CO_2, provides lifting energy during flowback through foam expansion once fracture pressure is released. Moreover, foamed acidizing enhances treatment effectiveness by improving acid placement (due to better fluid loss control and increased viscosity), increasing acid efficiency, and assisting formation cleanup (Briscoe, H. R. 1979); (Ford, W. G. F. 1981).

For over a decade, successful applications of foamed CO_2 acid treatments in the region have been well documented (Malik, A. R., 2014);(Sanchez, M 2015);(Rahim, Z. Al-Kanaan 2018);(Abel, J. T 2019); (Gromakovskii, D 2019);(Almutary, T 2021);(Liu, P 2022);(Al-Muhanna, D 2022); (Basset, M 2024); (Asel, S 2024). Consistently, these case studies report positive outcomes such as higher production rates (compared to both conventional and N_2-energized treatments), shorter post-job flowback times, faster formation cleanup, and reduced water consumption per treatment.

The rheological properties of foamed acid, especially its high viscosity, enable deeper and more effective fracture propagation and acid etching within the carbonate formation (Briscoe, H. R. 1979); (Ford, W. G. F. 1981); (Thompson, K. E 1993); (Talabani, S 1999). The foam structure acts as a diverter, temporarily blocking larger pores or natural fractures encountered near the wellbore, which promotes more uniform fluid distribution across the target zone. Without this, less permeable sections of the rock might be bypassed by the acid (Thompson, K. E 1993); (Talabani, S 1999). As a result, longer and more conductive fracture networks are created compared to conventional acid treatments, leading to substantial increases in production.

Beyond improving the performance of reservoir, foamed CO_2 acid fracturing offers notable operational and environmental benefits documented (Malik, A. R., 2014);(Sanchez, M 2015);(Rahim, Z. Al-Kanaan 2018);(Abel, J. T 2019);(Gromakovskii, D 2019);(Almutary, T 2021);(Liu, P 2022);(Al-Muhanna, D 2022); (Basset, M 2024);(Asel, S 2024). The reduced water use per treatment minimizes the logistical challenges associated with sourcing, transporting, and handling water. This is especially a valuable advantage in the fresh water scarce region of the Middle East. Faster cleanup and shorter flowback time enable quicker production startup of the well and reduce flaring time. Additionally, utilizing CO_2 gas (often captured from industrial processes) provides an opportunity for carbon utilization and storage within geological formations. This approach aligns with industry goals to lower emissions and water usage, supporting sustainability efforts in the Middle Eastern region.

Overview of the Foamed CO2 Acid Fracturing Operation

The equipment requirements for foamed CO_2 acid fracturing are similar to those of conventional acid fracturing treatments, with the key difference being the additional of specialized equipment to safely store, handle, and inject CO_2 in its liquid phase. Safety precautions and careful operational controls are required to manage and reduce the potential risks of handling high-pressure liquid CO_2, at the surface of the wellsite. Figure 1 shows a typical layout at the wellsite. The equipment is split into two main areas: one for the standard acid fracturing setup, and another dedicated to CO_2 handling. A separate high-pressure line, dedicated to pumping CO_2, downhole is rigged up and connected directly to the wellhead, allowing better control and monitor of CO_2. Mixing of liquid CO_2, and other fluids (acid, gel, and diverter) occurs at the wellhead.

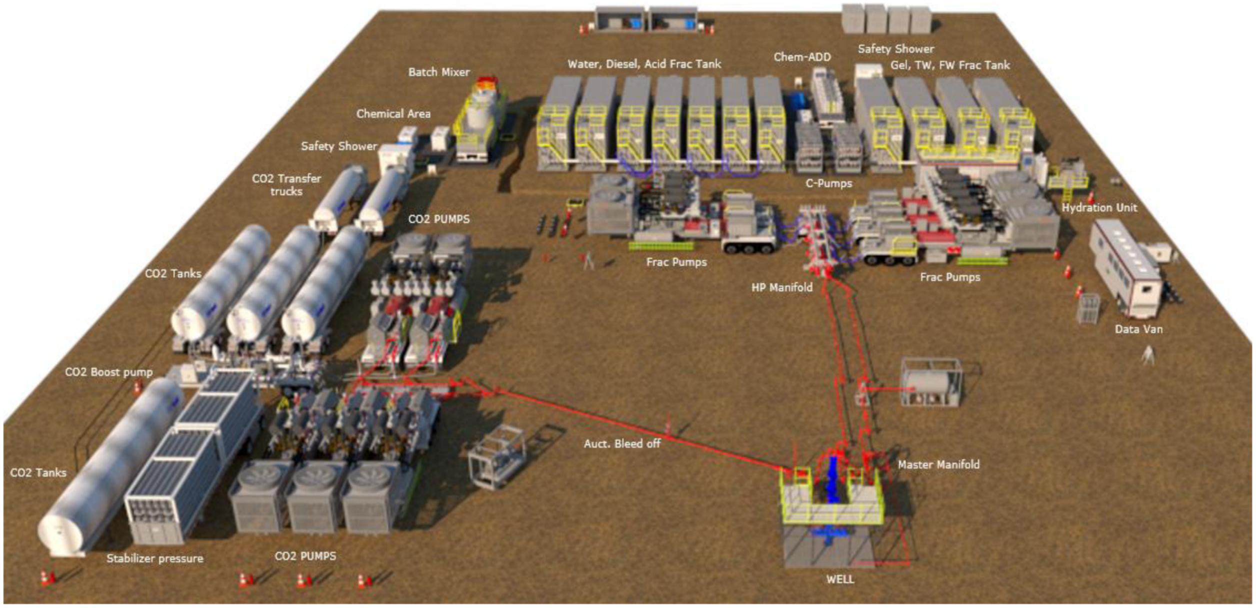

Figure 1—Typical equipment layout for foamed CO_2, acid fracturing treatment.

Once transferred from its transport bulker, liquid CO_2, is stored in pressure vessels at 100 psi or up, depending on weather conditions. Each vessel is equipped with its own pressure-relief valve to safely discharge CO_2, to atmosphere when required, releasing pressure buildup in the vessel. These vessels are then connected via CO_2,-rated hoses to a boost centrifugal pump that will increase pressure when feeding liquid CO_2, to high-pressure fracturing pumps (HP pump). Suction manifold of the HP pump does not utilize any vitaulic or clamp connection, which could cause CO_2 leakage. This is shown in Figure 2.

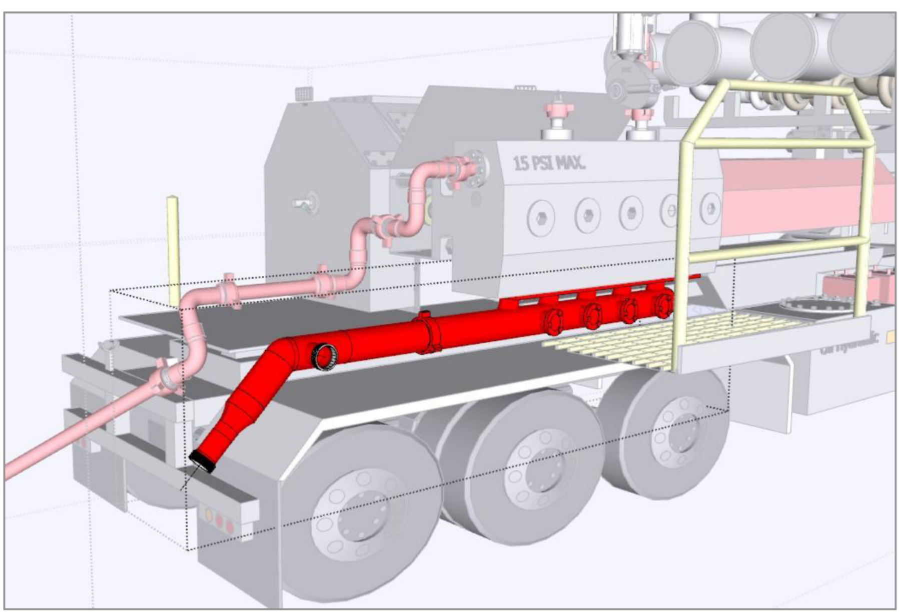

Figure 2—Suction manifold of CO_2, high-pressure pump.

Every fluid end of the pump is configured to have its own CO_2 venting setup (a bleed-off assembly with choke to discharge it to the atmosphere) operated by remote actuated valve as shown in Figure 3. In the discharge line of the fluid end, isolation and dart check valves are installed before lateral connecting to the main CO_2 high pressure line. In the high-pressure line, additional devices are installed to control pressure and ensure safe operation: venting valve, bleed-off assembly, dart check valve and isolation valve.

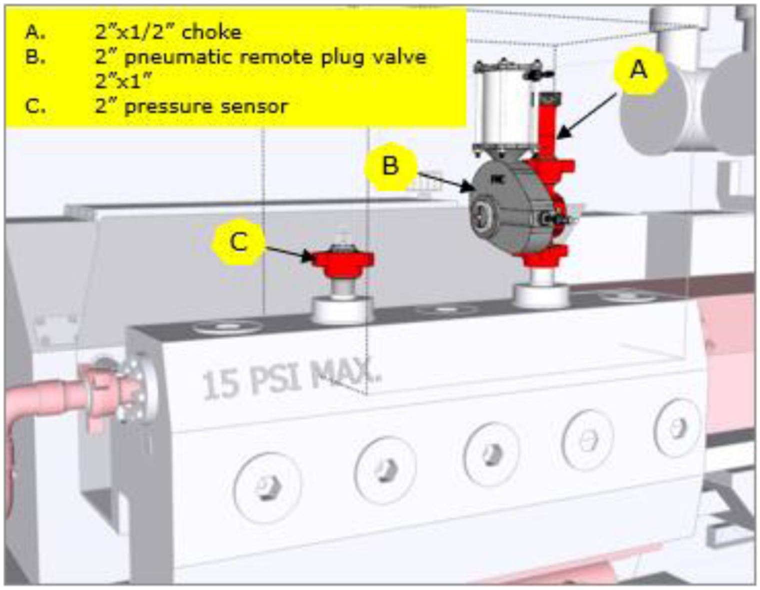

Figure 3—CO_2, venting setup in pump's fluid end.

Before starting the operation, the CO_2 high-pressure line is pressure tested using glycol. It is critical that CO_2, HP pump and treating line never contains water, as residual water react with CO_2 to produce ice, which can block the line and cause dangerous pressure buildup. Once acceptable pressure result is obtained, glycol is drained into a catch tank. The system is then purged with vapor CO_2 to remove any remaining moisture, ensuring the line is dry and safe for operation.

Prior to pumping liquid CO_2, into the well, the high-pressure CO_2 pumps and lines are cooled with liquid CO_2 itself. Once cooled, the pumps and lines are primed with liquid CO_2, and injection into the wellhead proceeds according to the approved pumping schedule. Throughout the operation, the boost pump needs to supply pressure between 200 and 300 psi to the suction of HP pumps. This is to ensure that CO_2, is pumped in its liquid state.

Case Study: Multistage Fracturing of Well A

This case study highlights the successful application of foamed CO_2 acid fracturing in a carbonate formation. Well A was drilled horizontally to a total measured depth of 17,901 ft, targeting a large, laterally extensive zone. The well was completed with a 4.5-inch × 13.5 lb/ft tubing, a 4.5-inch × 7-inch liner hanger, and lower completion containing multistage fracturing sleeves activated via ball drop in a cemented liner. The lateral length was 4,000 ft, with reservoir pressure estimated at about 4,500 psi and bottomhole temperature around 300°F.

Six fracturing stages were designed to maximize gas extraction, with strategic placement of frac ports in zones having at least 5% porosity. Each stage was designed with acid volume of 300 to 500 gallons per foot of pay zone, maintaining CO_2, foam quality of 60%. The treatment design employed a cyclic sequence of a crosslinked gel (PAD), 26% HCl acid, and diverting agents (both chemical and solid diverters), similar to conventional treatments but with enhanced operational control for CO_2 handling. Unlike conventional acid job, emulsified acid (a mixture of acid and diesel) is not utilized in the foamed acid treatment because it loses stability upon contact with CO_2. Table 1 shows pumping schedule for one of the stages in Well A.

Table 1—Treatment schedule for one of the stages in Well A.

Step #	Step Name	Fluid Name	Fluid Volume	Fluid Rate	CO2 Rate	Total Pump Rate	CO2 Volume	BH Foam Quality	Pump Time	Total Stage Volume
			(bbl)	(bbl/min)	(bbl/min)	(bbl/min)	(Tonne)	(%)	(min)	(gal)
1	Cool Down Stage	Treated Water	75	8.0	0.0	8.0	0.0	0	9.4	3,150
2	Acid-0	CSA HCl 26%	48	12.0	0.0	12.0	0.0	0	4.0	2,000
3	*Acid Displcmnt	Treated Water	190	12.0	0.0	12.0	0.0	0	15.9	8,000
4	Pad-1	40# LpH Sirocco (CO2)	29	6.0	8.7	14.7	6.7	60	4.8	2,939
5	Acid-1	CSA HCl 26% (CO2)	38	6.0	8.7	14.7	9.0	60	6.3	3,919
6	Diversion-1	Lo-Gard (CO2)	29	8.0	11.6	19.6	6.7	60	3.6	2,939
7	Pad-2	40# LpH Sirocco (CO2)	29	8.0	11.6	19.6	6.7	60	3.6	2,939
8	Acid-2	CSA HCl 26% (CO2)	52	8.0	11.6	19.6	12.3	60	6.5	5,389
9	Diversion-2	Lo-Gard (CO2)	29	8.0	11.6	19.6	6.7	60	3.6	2,939
10	Pad-3	40# LpH Sirocco (CO2)	33	10.8	15.7	26.5	7.8	60	3.1	3,429
11	Acid-3	CSA HCl 26% (CO2)	57	10.8	15.7	26.5	13.4	60	5.3	5,879
12	Spacer	Linear Gel	4	8.0	0.0	8.0	0.0	0	0.5	168
13	Diversion-3	NWB Diverter	4	8.0	0.0	8.0	0.0	0	0.5	168
14	Spacer	Linear Gel	8	8.0	0.0	8.0	0.0	0	1.0	336
15	Pad-4	40# LpH Sirocco (CO2)	38	12.0	17.4	29.4	9.0	60	3.2	3,919
16	Acid-4	CSA HCl 26% (CO2)	62	12.0	17.4	29.4	14.5	60	5.2	6,369
17	Diversion-4	Lo-Gard (CO2)	29	12.0	17.4	29.4	6.7	60	2.4	2,939
18	Pad-5	40# LpH Sirocco (CO2)	43	12.0	17.4	29.4	10.1	60	3.6	4,409
19	Acid-5	CSA HCl 26% (CO2)	62	12.0	17.4	29.4	14.5	60	5.2	6,369
20	Diversion-5	Lo-Gard (CO2)	29	12.0	17.4	29.4	6.7	60	2.4	2,939
21	Pad-6	40# LpH Sirocco (CO2)	43	14.0	20.3	34.3	10.1	60	3.1	4,409
22	Acid-6	CSA HCl 26% (CO2)	62	14.0	20.3	34.3	14.5	60	4.4	6,369
23	Over Flush (CO2)	Treated Water (CO2)	50	12.0	17.4	29.4	11.7	60	4.2	5,144
24	Flush (CO2)	Treated Water (CO2)	67	12.0	17.4	29.4	15.7	60	5.6	6,859
25	Flush no CO2	Treated Water	12	30.0	0.0	30.0	0.0	0	0.4	500

In every stage, prior to the main treatment, injection test was conducted to verity well's injectivity. Once the injectivity of the stage was confirmed, the preparation of materials commenced. This involved transferring raw acid into storage tanks, where it was mixed with the necessary additives. Additionally, gel and treated water were mixed, and liquid CO_2 was transferred from its transportation vessels. With all required resources arriving promptly at the wellsite, and without any delays, the time required between the execution of the main stages was approximately 24 hours. Figure 3a presents the chart illustrating the main treatment process across the stages. Throughout the 6 stages of foamed CO2 treatment of Well A, the following materials were pumped in these quantities: 101,000 gallons of mixed acid, 72,000 gallons of crosslinked gel, 25,000 gallons of chemical diverter, 4,300 pounds of solid diverter, and 1,300 tons of liquid CO2 (covering both cooling down and priming up).

Once the final stage of the treatment was completed, the well was closed to ensure the acids from the last stage fully reacted with the formation rock. After six hours, the well was opened for flowback. The flowback started with a small choke, gradually increased to bigger size during the bean up process. The maximum gas flow rate achieved during post-treatment flowback was more than 10 mmscf/d with a wellhead pressure of around 2,700 psi. The flowback period took about five days, during which the basic sediment and water (BS&W) content decreased to 8% and the CO2 concentration dropped to 2%.

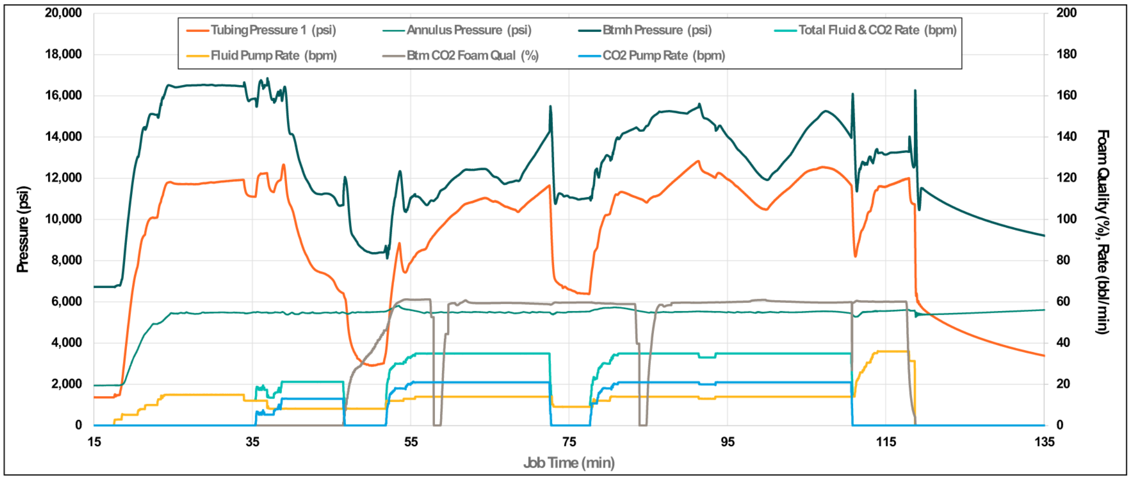

Figure 3a—Treatment chart for one of the foamed acid fracturing stages in Well A.

Operational Efficiency and HSE

All stages were completed with zero incidents related to health, safety, and environment, emphasizing commitment to HSE excellence throughout the operation. The fracturing operation on Well A was executed with high service quality, ensuring adherence to all technical and operational standards. With effective planning and execution, the operation progressed efficiently within the planned schedule.

Throughout each stage, all mixed fluids met the strict QA/QC requirements, ensuring optimal performance and reliability. Additionally, the equipment available on the wellsite was capable of delivering the necessary horsepower for all six stages, supporting a smooth and continuous operation. Emphasizing safety, quality, and equipment readiness, these factors contributed significantly to the success and safety of the fracturing operation for Well A.

Productivity Analysis

As it can be seen from Fig. 4, CO2 fracturing was performed in the most depleted wells throughout the campaign. This also led to wellhead flowing pressure to be lower than with the other two groups. Overall, over 90 wells were compared within the study including 30 wells stimulated with CO2 foam fracturing.

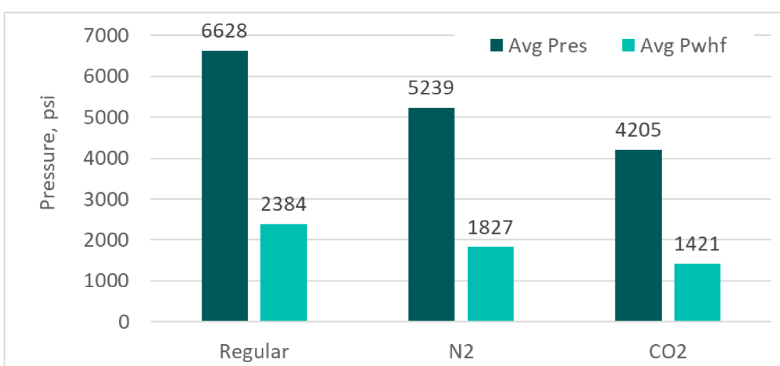

Figure 4—Reservoir Pressure (Pres) and wellhead flowing pressure (Pwhf) average by groups of wells – regular acid fracturing, nitrified and implementing CO2-based foam

Further, each group was split into 2 subgroups: single stage wells and wells with multiple stages – either vertical or horizontal completions. Fig. 5 shows comparison of the gas flowback rates – normalized with the average performance of regular fracturing in single stage completion. Initial flowback was performed consistently for each of the groups and lasted for several days. Highest rates were picked out of the flowback period. One can see a similarity between nitrified acid systems and regular operations (despite of more depleted wells in case of nitrogen) and 5% reduction of the initial flowback rates in case of carbon-dioxide foam implementation. Again, this reduction is mainly due to severe depletion of the formations selected for the CO2 fracturing. When comparing the multistage wells, one can see the increase in productivity by at least 20% in each group and higher increase (36% Vs 20%) in case of carbon dioxide use in fracturing system. This is mainly due to the fact that carbon dioxide Is trapped inside formation and stores additional energy for longer period of time, releasing it effectively and helping flowing back the well despite of lengthy period required to complete the multistage operations. Nitrified acid fracturing is used mainly for the last or last two stages per well because it cannot last effectively for longer periods and would not be helpful in case of multistage wells. This is definite advantage of CO2 foam fracturing compared to N2 fracturing implementation.

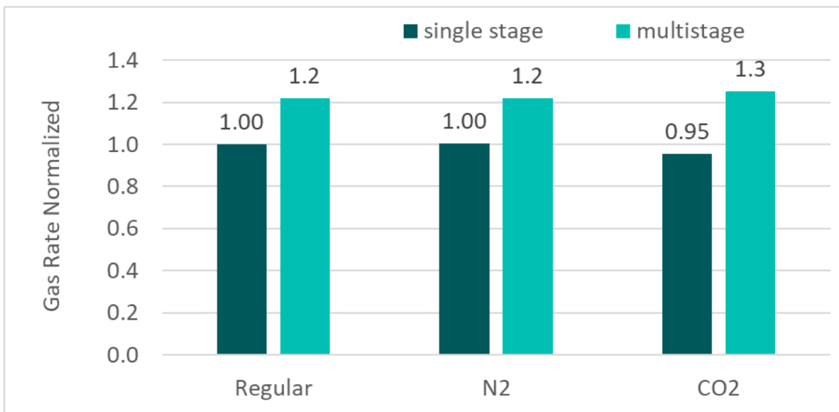

Figure 5—Normalized gas rate comparison between regular, nitrified and CO2-based foam fracturing for both single stage and multiple stage well completions

Fig. 6 provides perhaps the most representative comparison between the fracturing techniques when it comes to fracture quality evaluation. Gas productivity index (PI) was calculated from initial flowback taking consistent measurement of the rate and pressure drawdown. Since there was significant presence of liquid during original flowback, authors used an integrated criteria for the drawdown (somewhere in between pure liquid and pure gas, as both phases where present within the fracture and the reservoir for the duration of the initial flowback). Thus, instead of using ($P_{res} - P_{wbf}$) as for liquid or using (P^2_{res} - P^2_{wbf}) as for gases for the drawdown calculation, the power of 3/2 was used consistently throughout analyses that demonstrated better matching with reality and productivity history. $PI = Q_{gas} / (P^{3/2}_{res} - P^{3/2}_{wbf})$. One can observe a significant advantage of the productivity when CO_2 foam fracturing is implemented. The reason for it is not only additional energy CO_2 stores to improve flowback, but also because the total volume injected into formation was considerably higher in these well. Acid amount was comparable between each acid fracturing operations including nitrified and carbon dioxide foam fracturing, meaning the rock dissolution capacity was comparable. However, when we take into account the gaseous phase – it represents significant amount, especially in case of CO_2 fracturing. Because foam quality of CO2 was 55-60%, meaning that the overall slurry injected into formation was more than doubled when compared to conventional fracturing stages. Eventhough the acid volume was the amount of rock dissolved was comparable, the length of the fracture propagation was substantially higher in case of CO_2 operations which brings significant benefit for the tight reservoir which was the case in our study. Nitrified fluids had 10-15% foam quality, which places them closer to conventional operations when it comes to generated fracture geometry when compared to CO_2 fracturing.

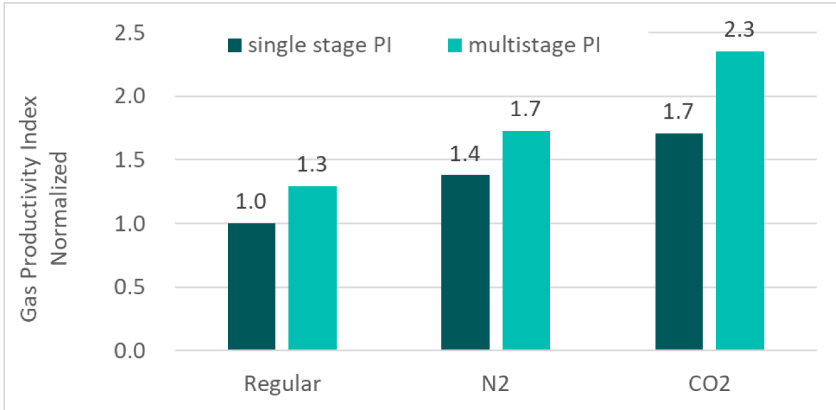

Figure 6—Normalized gas productivity index comparison between regular, nitrified and CO2-based foam fracturing for both single stage and multiple stage well completions

Conclusion

Carbon dioxide -based foam acid fracturing campaign was performed in more than 30 wells. Operational efficiency and safety were at the highest level ensuring the sustainable implementation of this technique and allowing for representative comparison on gas productivity of the wells. Three groups of wells were used in this study including conventional acid fracturing, nitrified acid and CO2 -based foam. Further each group was subdivided into single-stage wells and multistage stimulated wells. Following conclusions can be made based on the study:

Despite the fact that CO2 -based system was implemented in the most depleted wells, it allowed to maintain gas production rate during post-fracturing flowback. Nitrified fluids were implemented in moderately depleted formations and also helped to maintain flowback rates at high levels. Advantage of CO2 fracturing system was obvious in case of multistage wells implementation when compared to conventional method and nitrified.

When the groups were compared in term of productivity index by normalizing gas rate with the drawdown values, advantage of CO2-based foam fracturing reached as high as 70% for the single stage operations when compared to regular acid fracturing (while nitrified fracturing was 40% more effective than regular) and as high as 130% for the multistage operations (when nitrified fluids reached 73% advantage over conventional method).

Thus, CO2 foam fracturing is highly recommended for the depleted and sensitive gas formations provided it is executed properly and safely, some critical details and recommendations listed in the paper.

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